



Quick Start Guide

Cytek® Guava® InCyte™ Assay

Acquisition and analysis on the Guava easyCyte™ and Guava easyCyte™ HT systems For

Research Use Only. Not for use in diagnostic procedures.

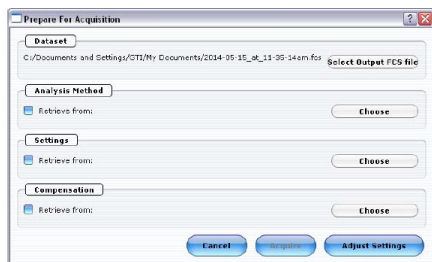
This quick start guide provides basic instructions on acquiring samples and analyzing data using the Guava InCyte software module. For detailed information, refer to the appropriate Guava easyCyte System User's Guide.

Acquiring Samples

Acquiring Samples on the Guava® easyCyte™ Systems

1. Select **InCyte** from the list of Favorites.
2. Click the **Start New Data Session File** button in the acquisition tool bar.

A dialog appears, allowing you to select a file name, Analysis Method, and instrument and compensation settings.



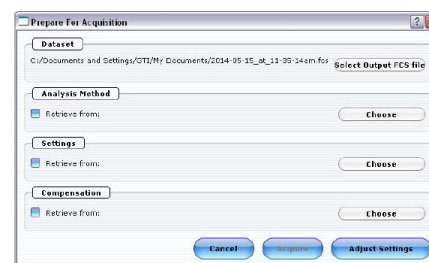
3. (Optional) Select the following for the acquisition.
 - a. Click **Select Output FCS File** to change the default FCS data file name and location.
 - b. Click **Choose** under **Analysis Method**, **Settings**, or **Compensation** to select the corresponding file.
You can obtain the Method or settings from an FCS file, or a Method (.gsy) or settings (.gst) file, respectively. If you do not want to use an existing Method, a default Method will be created. You can modify it during acquisition or analysis.
4. Click **Adjust Settings**.
If you retrieved instrument settings and you wish to skip the adjust settings step, click **Acquire**.
5. Mix a stained negative sample for the adjust settings step and load it on the instrument.
6. Click **OK** in the load sample dialog.
7. Fine tune the instrument settings:
 - Use the Miscellaneous controls to set the flow rate, and select the parameter measurements to acquire (area, width, time).

- Set up the plots.
- Use the gain controls to set and adjust the threshold and adjust the gains to distribute the population(s).
- Use the compensation controls to adjust the fluorescence compensation (can also be performed during analysis).
- You can set regions and gates for acquisition. To set a count gate, create a region(s), define the gate, then select it from **Count Gate** menu under **Sample Info**.

8. Click the **Next Step** button when you are finished adjusting settings.
9. Mix and load the first sample to acquire.
10. Enter the number of events to acquire and a sample ID, then click the **Acquire Next Sample** button.
11. When acquisition is complete, click the **Save and Close Current Sample** button.
12. Repeat step 5 ("Mix a stained..." through step 11 ("When acquisition is...") for the remaining samples.
13. Click the **Stop and Close Session File** button after the last sample.

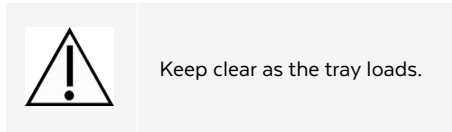
Acquiring Samples on the Guava® easyCyte™ HT Systems

1. Select **InCyte** from the list of Favorites.
2. Click the **Edit Worklist** button to open WorkList Editor and define the parameters for the worklist. If you have a saved worklist, click **Open** in the WorkList Editor and select the worklist. Proceed to step 4 ("(Optional) Select...").
3. Define the worklist parameters and click **Run Worklist**. A dialog appears allowing you to select a data file name, Analysis Method, and instrument and compensation settings.

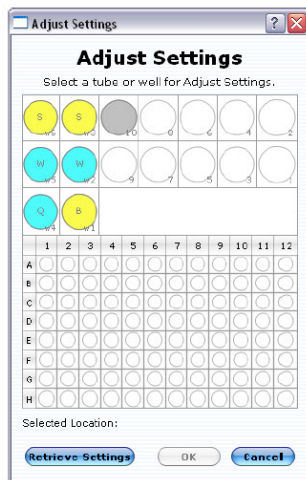


4. (Optional) Select the following for the acquisition.

- Click **Select Output FCS File** to change the default FCS data file name and location.
 - Click **Choose** under Analysis Method, Settings, or Compensation to select the corresponding file.
You can obtain the Method or settings from an FCS file, or a Method (.gsy) or settings (.gst) file, respectively. If you do not want to use an existing Method, a default Method will be created. You can modify it during acquisition or analysis.
- Click **Adjust Settings**.
If you retrieved instrument settings and you wish to skip the adjust settings step, click **Acquire**.
 - Follow the instruction in the dialog box that appears to load samples and cleaning tubes. Place the plate and/or tubes in the tray and click **OK**.



- A dialog box appears prompting you to select the sample for adjusting settings. Select the location of the sample you want to use to adjust settings and click **OK**.



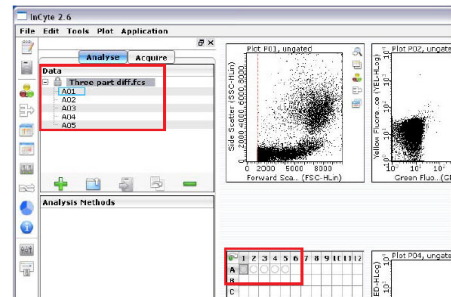
- If necessary, set up the plots.
- Fine tune the instrument settings:
 - Use the Miscellaneous controls to set the flow rate, and select the parameter measurements to acquire (area, width, time).
 - Set up the plots.
 - Use the gain controls to set and adjust the threshold and adjust the gains to distribute the population(s).
 - Use the compensation controls to adjust the fluorescence compensation (can also be performed during analysis).
 - You can set regions and gates for acquisition. To set a count gate, create a region(s), define the gate, then select it from the **Count Gate** menu under **Sample Info**.
- Click the **Next Step** button when you are finished adjusting settings.
- Click the **Resume Worklist** button to acquire all the samples in the worklist.

Analyzing Data

Data files acquired using a GuavaSoft™ module other than InCyte™ will not have an Analysis. However, when you open the file in InCyte, a default Method will be created.

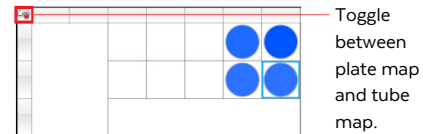
Analyzing Files Acquired Using InCyte™

- From the **Analyse** screen, choose **File > Open**. Select the FCS file and click **Open**.
The data for the first sample appears. Click the **+** to the left of the data file name in the **Data** pane to display a list of all samples contained in the file. Click any sample in the list to see the unanalyzed data for that sample in the plots.



If the data was already analyzed, click the Analysed Group (third pane in the control panel) to see the analysis gates and markers. Click the individual wells in the plate map to see analyzed data for that sample in the plots.

NOTE: If data was acquired on an easyCyte™ HT, click the icon in the upper-left corner of the plate map to toggle between a plate map and tube map.



- Customize the analysis using regions and gates. See “Applying Regions, Gates, and Statistics” on page 2.

Analyzing Files Acquired Using Other GuavaSoft™ Modules

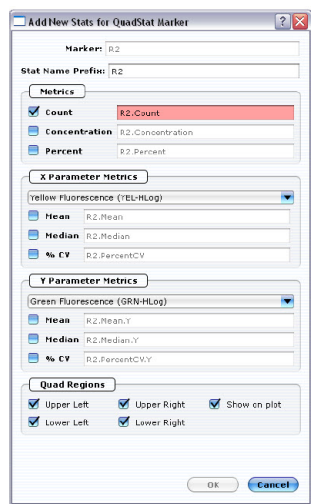
When you open a file acquired using a GuavaSoft™ module other than InCyte™, a default Analysis Method is created. You can modify the Method or open an existing Method, then drag it to the Analysed Group. For more information on analyzing non-InCyte acquired files, refer to the appropriate Guava® easyCyte™ System User’s Guide.

Applying Regions, Gates, and Statistics

- Click the **New Region** icon in the plot tool bar and select a region type from the menu.
- Enter a region name or leave the default, and click **OK**. If you are creating a polygon, create the region and then enter the region name.
- To create a gate, click to select the region, then drag it to another plot. This applies the data within the region to the second plot and creates a gate defined with the “AND” operator. To create gates using OR or NOT, use the Gate List. See the appropriate easyCyte™ System User’s Guide for information.
- Enter a name for the gate and click **OK**.
- To create statistics, click the **New Stat Marker** icon in the plot tool bar, then select the type of stats you want. If you wish to get

stats on an existing region, select the region. If you wish to get stats on a new region not yet created, select the new stat marker, for example, select **New Quad Stat Marker**. This displays the Add New Stat Marker window (shown below), allowing you to define the type of stats (metrics) and create/adjust the new region in the plot.

- 6. Select the metrics and prefix for the stat name, then click **OK**.



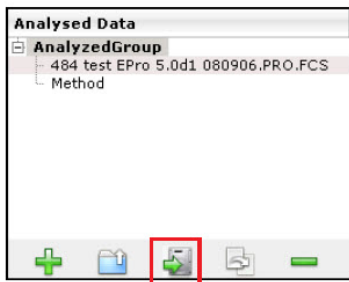
The stat is automatically applied to all samples in the data set. To see the results for a specific sample, place the cursor over the well in the plate map.

The **Current Run Stats** window appears displaying a list of all the stats associated with the Method. The values displayed correspond to the sample well selected in the plate map. Click any well to see the stats for that sample.

Current Run Stats						
Stat Name	Value	Metric	Units	Marker Name	Quad	Parameter
R2.Count.UL	637	Count		R2	Upper Left	---
R2.Count.UR	3	Count		R2	Upper Right	---
R2.Count.LL	3166	Count		R2	Lower Left	---
R2.Count.LR	3	Count		R2	Lower Right	---

Saving the Analysis Session

- 1. To save the analysis, click on the **Analysed Group** in the **Analysed Data** pane to select it, then click the **Save Analysed Group** icon.



- 2. Navigate to the location, enter a name, and click **Save**.
The Analysed Group will be saved as an FCS file and will contain the data and the Method (regions, gates, metrics, and stats).

For detailed information on the various InCyte™ features, refer to the appropriate easyCyte™ System User's Guide. Some InCyte features are listed below.

- Heatmapping – provides a colorimetric representation of results in the platemap and allows you to simultaneously compare up to six different results (analyses) for samples within a dataset.
- IC50/EC50 applications – use to obtain information on the kinetic characteristics of a compound.
- Compensation – correct for fluorescence spillover before or after acquisition, or use the semi-automated feature to perform compensation during analysis.
- Statistics – customize statistics, create expression stats to a apply dilution factor to get stats on diluted samples.
- Instrument settings – settings are saved with each FCS file, and can also be saved to separate settings files, and applied to future acquisitions.
- Data pooling – combine data from multiple samples into a single sample when analyzing rare events.

Technical Support

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